

20. Does the system of congruences $x \equiv 1 \pmod{8}$, $x \equiv 3 \pmod{9}$, and $x \equiv 2 \pmod{12}$ have any simultaneous solutions?

10. How many incongruent solutions are there to the congruence $x^5 + x - 6 \equiv 0 \pmod{144}$?

8. Find an inverse modulo 5 of each of these matrices.

a) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ b) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ c) $\begin{pmatrix} 2 & 2 \\ 1 & 2 \end{pmatrix}$

7. What is the remainder when $18!$ is divided by 437?

2. Show that 45 is a pseudoprime to the bases 17 and 19.

17. Find a Carmichael number of the form $7 \cdot 23 \cdot q$, where q is an odd prime other than $q = 41$, or show that there are no others.

19. Use Euler's theorem to find the last digit in the decimal expansion of 7^{1000} .

3. Show that $\phi(5186) = \phi(5187) = \phi(5188)$.

1. Find the primes p and q if $n = pq = 14,647$ and $\phi(n) = 14,400$.